Mining is the distributed computational review process performed on each block of data in a blockchain, which allows a consensus to be achieved despite each party not knowing or trusting the other.

Let us take the example of Bitcoin to understand this better. In order to conduct a transaction, a Bitcoin user has to sign it with his private key or seed. This helps prove that the transaction has been initiated by the right person, and also prevents third parties from altering the transaction in any way. After this, the transaction is confirmed through a process of mining and then included in the blockchain.

According to Bitcoin, "To be confirmed, transactions must be packed in a block that fits very strict cryptographic rules that are verified by the network. These rules prevent previous blocks from being modified because doing so would invalidate all the following blocks.

"Mining also creates the equivalent of a competitive lottery that prevents any individual from easily adding new blocks consecutively in the block chain. This way, no individuals can control what is included in the blockchain or replace parts of it to roll back their own spends."

Mining makes computer hardware do mathematical calculations for the Bitcoin network to confirm transactions and increase security. Any computer across the world can be a miner. As a reward for sharing their computational power, miners get a fee for each transaction confirmed by them, along with newly-created bitcoins. The reward depends on the amount of mining done. There are strict rules and procedures for this, and mining is not to be thought of as an easy way to make money.

Blockchain mining hardware. The blockchain mining process is mostly software based. In reality, anybody can become a miner using their personal computer. However, the speed and reliability demanded by upcoming mining protocols pose an opportunity for hardware innovation. We now have industrial-grade mining hardware (servers in data centres) and specialised application-specific integrated circuits (ASICs) for mining.

Antminer, Avalon6, SP20 Jackson and 21 Bitcoin Chip are some well-known ASICs for mining. Experts say you have to look at hash rate, efficiency and price when selecting a mining chip. Hash rate is the number of hashes per second that the Bitcoin miner can make. Usually, if the hash rate is more, the price will be higher. So you have to balance it with efficiency, which measures the electricity usage. This is important because miners usually consume a lot of power. Remember, we told you it is not an easy way to make money!

21 Inc. has come up with something more than ASICs. 21 Bitcoin computer is claimed to be the first computer with native hardware and software support for Bitcoin protocol. According to the company, it has the hardware to mine a stream of small amounts of Bitcoins for development purposes, and the software to make that Bitcoin useful for buying and selling digital goods. It can be used by developers and individual miners to create Bitcoin based apps, services and devices.

Embedded blockchain mining is a very nascent idea, which involves embedding mining chips into different kinds of Internet-connected devices. Last year, 21 Inc. revealed its plans to develop a chip (called 21 BitShare) that can be embedded into mobile phones, enabling them to silently mine in the background.

Another innovation from 2015, by Patric Lanhed and Juanjo Tara, is a chip that can be implanted in your hands for Bitcoin bio-payment. They have gone a bit overboard, but it might be a glimpse of what